

Amjad S Althabteh

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Education

B.S in Computer Science GPA: 3.55/4.00 **Wayne State University** *Detroit* **2026**

Experience

SWE Intern	Immersalab Graphics	<i>Detroit, MI</i>	Sep 2025
- Architected a real-time rendering engine in C++20 and OpenGL, focusing on GPU-driven rendering. - Robust internal UI and asset-authoring tools built on JSON-driven pipelines, greatly expanding the flexibility of content creation. Data-driven rendering optimizations to allow runtime analysis and AI guided heuristics to balance visual quality, GPU performance. - Modular rendering subsystems with clear ownership boundaries to support iteration and feature growth - GPU-side profiling and frame-timing instrumentation to guide optimization and debugging.			
Unreal, C++	Indie Game AAA	<i>Ann Arbor, MI</i>	Apr 2025 - Aug 2025
- Wrote gameplay systems using (C++ 20 and Visual Scripting), connecting animation states, environmental triggers, and narrative logic to create seamless transitions between cinematic moments. - Created 3d environments and characters from scratch, rigging for animation, and enhanced each scene. - Produced a 90+ second real-time cinematic using MetaHumans, Control Rig, and Sequencer designing the flow. - Dynamic animation-control layer using C++ and Animation Blueprints, making smooth blending across all aspects. - Implemented event-driven state transitions to synchronize gameplay logic, animation, and cinematic systems			

Software Development Intern	Rare Munchies	<i>Ann Arbor, MI</i>	Sep 2023 - Apr 2024
- Integrated UI components using TypeScript, improving page transitions and interaction flow to enhance user experience. - Identified and resolved cross-browser compatibility issues, reducing debugging time and unblocking delivery - Resolved cross-browser compatibility issues, reducing debugging overhead and stabilizing releases			

Projects

Glint 3D – C++ Automated 3D Rendering Engine

[GlintRender](#)

- Rendering system built on **C++17/OpenGL rendering pipeline with dynamic GPU workload** and lighting optimization.
- Designed a cross-platform architecture with runtime scene analysis to maintain stable performance in complex scenes
- Set up a **modular engine architecture** separating rendering, allowing faster iterations.
- Implemented a **custom memory pool and allocator system** using C++20 features to reduce allocations, + runtime performance.

UE -5.6 Cinematic “No One Answers Clip”

[no one answers clip](#)

- Built a AAA solo cinematic in Unreal Engine 5.6 using MetaHuman animation and cinematic cameras, and control; rigged anims.
- Used Niagara effects and environmental animation to tell the story visually and set the tone for No One Answers

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Developed a **C++ 20 debugging system** that performs automated stack-trace parsing.
- Implemented automated regression test generation and intelligent fix suggestions to reduce debugging time.
- Added **thread-safe logging** to capture errors and runtime events in multithreaded applications.
- Basic **memory tracking utilities** to help identify leaks and invalid allocations during debugging.

Incident Management Ops

- Built a real-time incident management platform with **FastAPI + React**, supporting collaboration rooms, async workflows, and Prometheus metrics for production observability
- Developed an end-to-end monitoring pipeline with automated alert ingestion, severity classification.

Skill

- **Programming Languages:** **C++ 11 - 26**, C#, C, Python 3.7 - 3.12, TypeScript, Java
- **Graphics & Engines:** OpenGL • **Unreal Engine 4 - 5.7** • Unity • • NX Design
- **Course Work:** Computer Graphics, Game Programming, **Software Engineering**, **Operating Systems**, Linear Algebra
- **Specialties:** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art